Week 5

COGS 108 SP 2026

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DI slides!

https://github.com/shreyamusini/cogs108_sp26_di_slides_shreya

```
% git clone https://github.com/shreyamusini/cogs108_sp26_di_slides_shreya.git
```

Due dates

- Project proposal: Wednesday (04/29)
- D4: Friday (05/01)

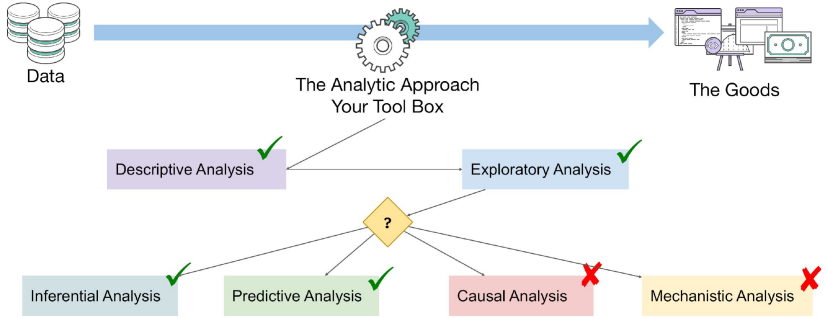
Project Proposal

- Due: Wednesday (04/29)
- Just make sure you’ve pushed your completed project proposal notebook to your github group repo by 11:59pm

Analysis styles: classify the goal

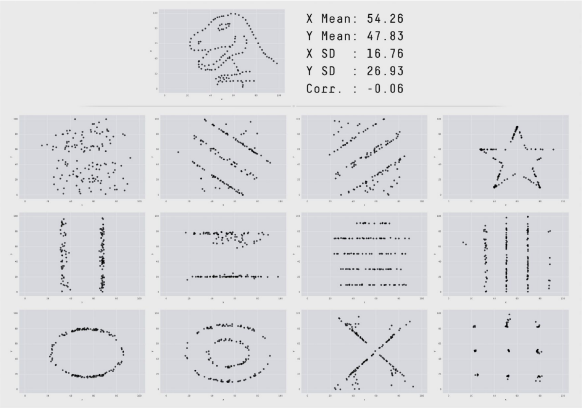
Focus on what the analysis is trying to do, not the dataset alone.

- Descriptive: summarize what is in the observed data.
- Exploratory: look for patterns and generate hypotheses.
- Inferential: estimate uncertainty and test claims about a population.
- Predictive: use past/observed data to forecast or classify new cases.
- Causal/mechanistic: ask what happens when one variable is changed.



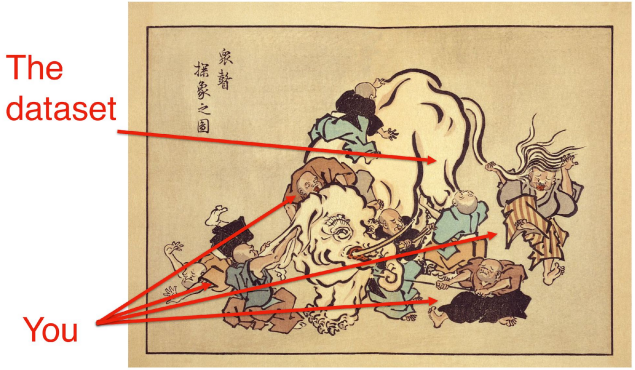
EDA mindset: one dataset, many views

Summary statistics can compress away important structure.



https://www.autodesk.com/research/publications/same-stats-different-graphs

A modern version of Anscombe's Quartet



Review prompts: What information is lost in a mean/SD? Why does EDA usually require multiple complementary plots?

Visualization quality checklist

A good plot helps a naive viewer understand the variable, units, and story.

Plot choice

Match the geometry to the task: distributions, trends over time, comparisons, relationships.

Labels & units

Replace raw column names with human-readable labels; add meaningful axis text.

Color discipline

Use color to encode information; avoid unnecessary palettes or inaccessible combinations.

Context

Use titles, annotations, legends/direct labels, and data source notes when helpful.

From code output to reader-friendly plots

Default plotting labels are often valid code, but poor communication.

```
import seaborn as sns
ax = sns.histplot(data=df,
x='stat_score')

# Then ask:
# What does the viewer need to know?
```

- What real-world construct does the column encode?
- What scale or units are being used?
- What does the count/density/frequency axis mean?
- Would a title or annotation prevent confusion?

Sampling and inference

Inference depends on whether the sample represents the target population.

Pearson's *r* :
linear correlation between two variables
takes values [-1,1]

Good sampling questions

Who is the target population? Who had a chance to be sampled? What groups are excluded or overrepresented?

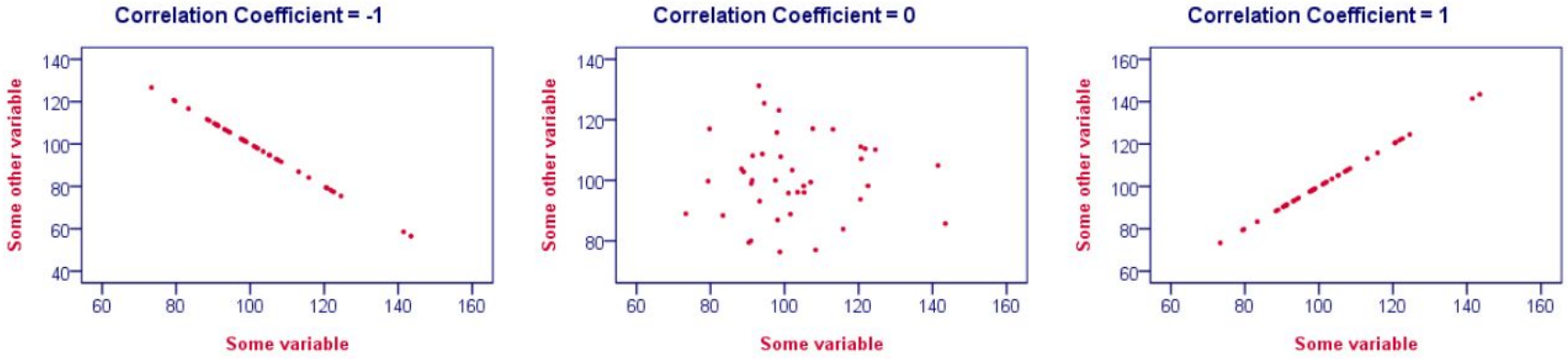
Bias patterns

Convenience samples, platform-specific samples, customer lists, interest-group audiences, and mismatch between sample frame and population.

Correlation and Pearson r

Pearson r describes linear association: direction and strength, not causal mechanism.

Correlation is how close the data are to being in a line…
BUT IT HAS NOTHING TO DO WITH THE SLOPE



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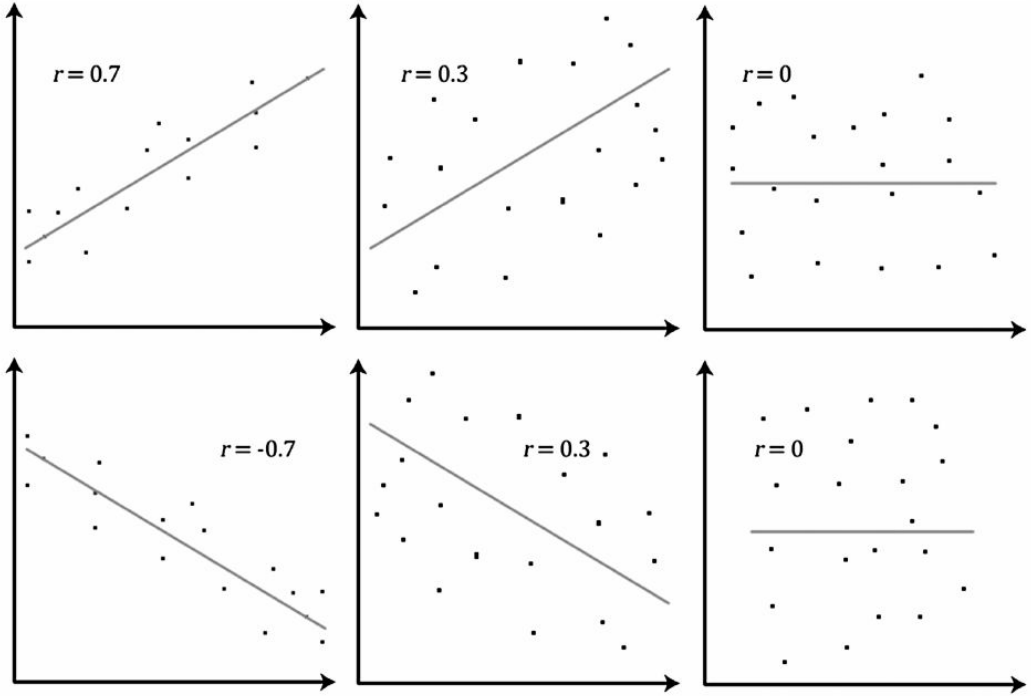
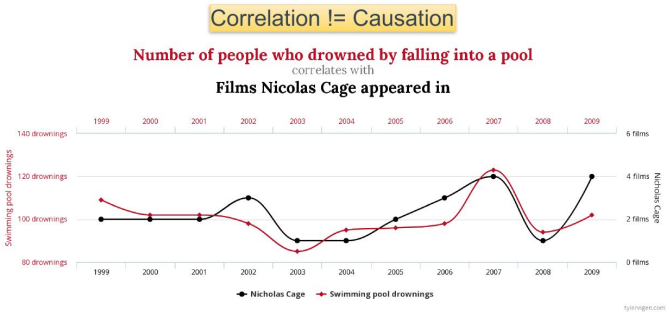


https://ascelibrary.org

Correlation is not causation

A relationship can exist for several non-causal reasons.



http://www.tylervigen.com/spurious-correlations

- Confounding: an unobserved third variable may drive both variables.
- Chance: random data can sometimes align.
- Shared time trend: both variables can rise or fall together for unrelated reasons.
- Reverse or bidirectional causality: direction may be ambiguous.

NHST and p-values

Null-hypothesis significance testing asks how surprising the observed result is under a null model.

p-value : the probability under the **null hypothesis** of getting measurements as extreme as the observed results by chance alone

THIS IS NOT TYPE 1 ERROR RATE

Null model

The reference world where the effect/difference being tested is absent or as specified.

p-value idea

Probability of observed-or-more -extreme results under the null model.

Review prompt: separate “statistical evidence” from “practical importance.”

Choosing comparison tests

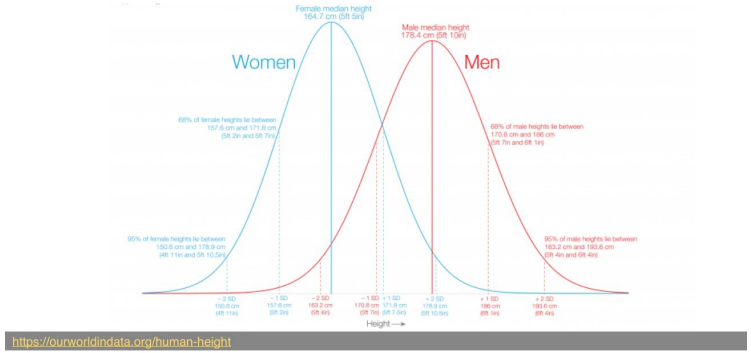
Start by identifying the study design and the measurement type.

Are groups independent?	Different people/items in each condition.
Are measurements paired/repeated?	Same unit measured more than once or matched pairs.
How many groups?	Two groups vs more than two conditions.
What assumptions look reasonable?	Continuity, approximate normality, variance similarity, and sample size.
Do assumptions fail badly?	Consider robust/non-parametric or simulation-based alternatives.

t-test validity: assumptions and traps

A t-test is not just a function call; the design and data shape matter.

Do the heights between males and females differ?



https://ourworldindata.org/human-height

- Continuous outcome variable, not categorical counts/labels.
- Approximately normal distributions, especially for small samples.
- Comparable variance across groups, or use a variant designed for unequal variance.
- Independence/paired structure handled correctly.
- A non-significant p-value is not an “invalid test” by itself.

Effect size vs p-value

Statistical detectability and real-world magnitude are different questions.

P-value

Evidence against a null model; sensitive to sample size and variability.

Effect size

Magnitude of a difference or association; connects results to practical meaning.

Interpretation

A complete interpretation considers evidence, magnitude, uncertainty, and context.

Geospatial data basics

Map-making uses geometry plus attributes; projection converts Earth to a flat display.

Point

A single location, such as an event, sensor, or address coordinate.

Line

A connected path, such as a road, border segment, or route.

Polygon

An area boundary, such as a county, state, zip code, or country.

Projection

A dimensional reduction from a curved Earth to a two-dimensional map.

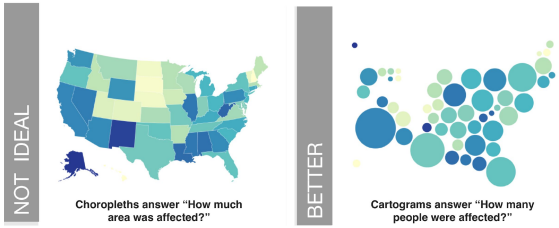
Tool topic: GeoPandas extends pandas-style workflows to spatial data frames.

Choropleths, normalization, and cartograms

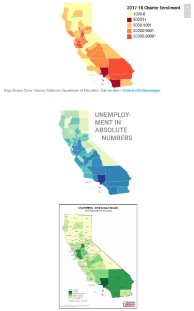
Before mapping, decide whether area or people should dominate the visual story.

- Choropleths color geographic areas; raw counts often reproduce population density.
- Normalize counts by population or area when the question asks about rates/intensity.
- Use light-to-dark schemes consistently when encoding low-to-high values.
- Cartograms can better represent how many people are affected.

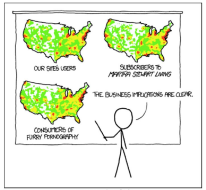
Cartograms should be considered when displaying how many people were affected



https://blog.datawrapper.de/choroplethmaps/



https://xkcd.com/1138/



If you don't normalize the statistic (divide by population or by area) choropleth maps can be misleading

Final self-check before the quiz

Visualization
Can I explain why a plot is readable or confusing?

Correlation
Can I estimate direction/strength and explain non-causal alternatives?

EDA
Can I explain why summary stats are insufficient alone?

NHST
Can I separate p-value, assumptions, effect size, and test choice?

Sampling
Can I identify mismatch between sample and target population?

Maps
Can I choose map types and explain normalization/projection/vector terms?

Quiz Time

NO Phone, NO Communication, No Notes

Access Code: code here

Your time to …

- Work on projects
- Finish D4
- Have fun...